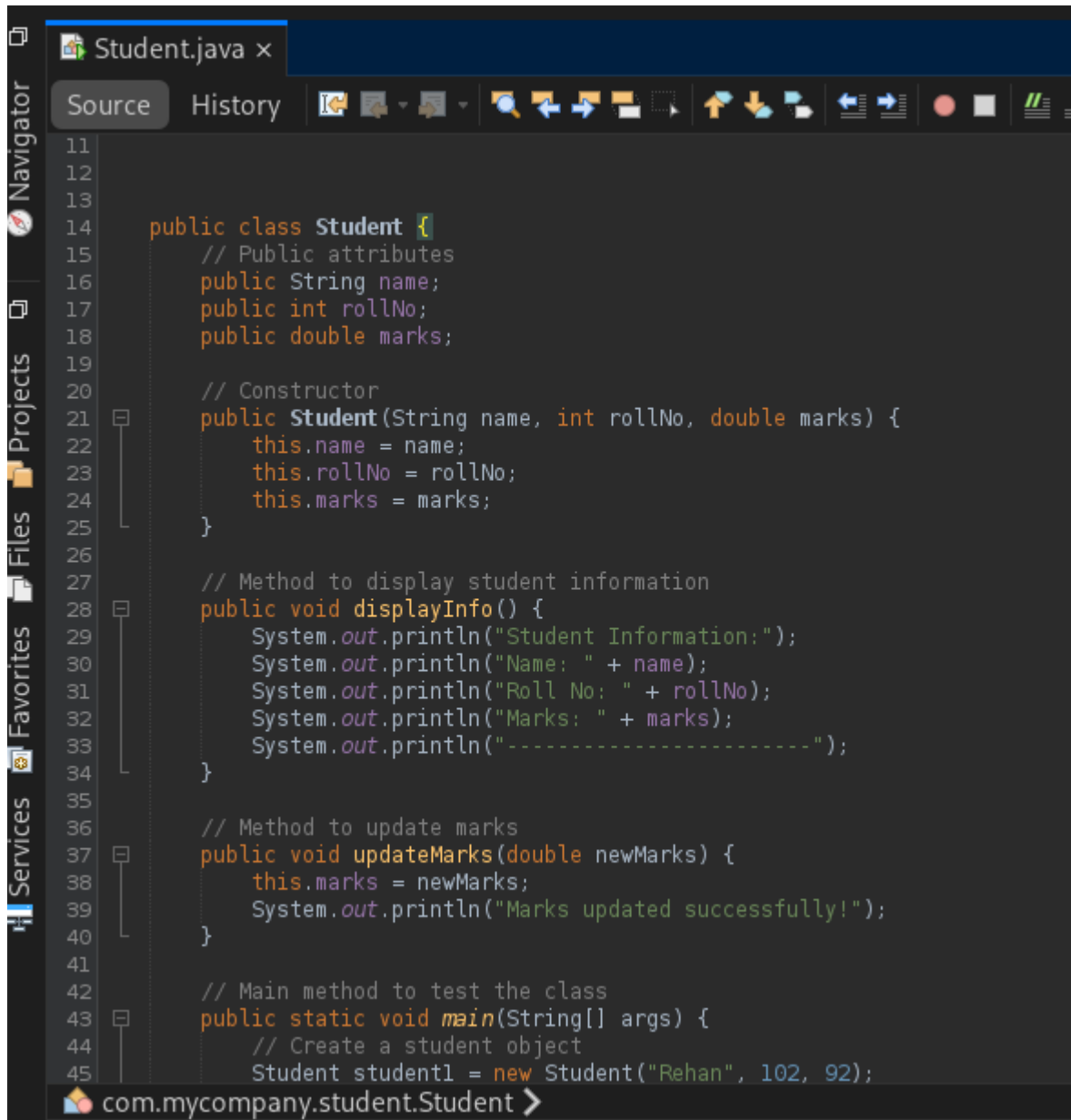


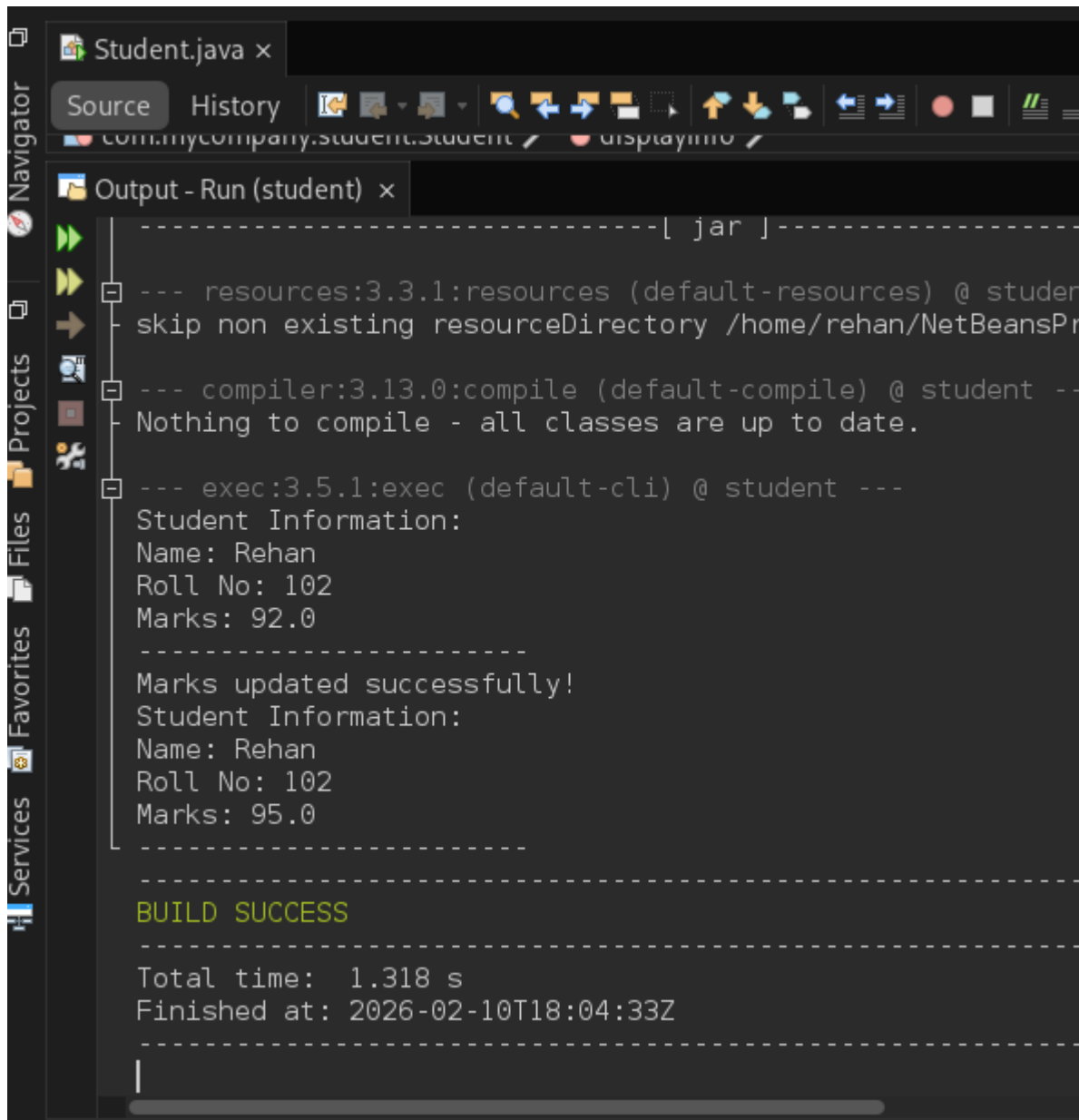
LAB#2

Task 1:



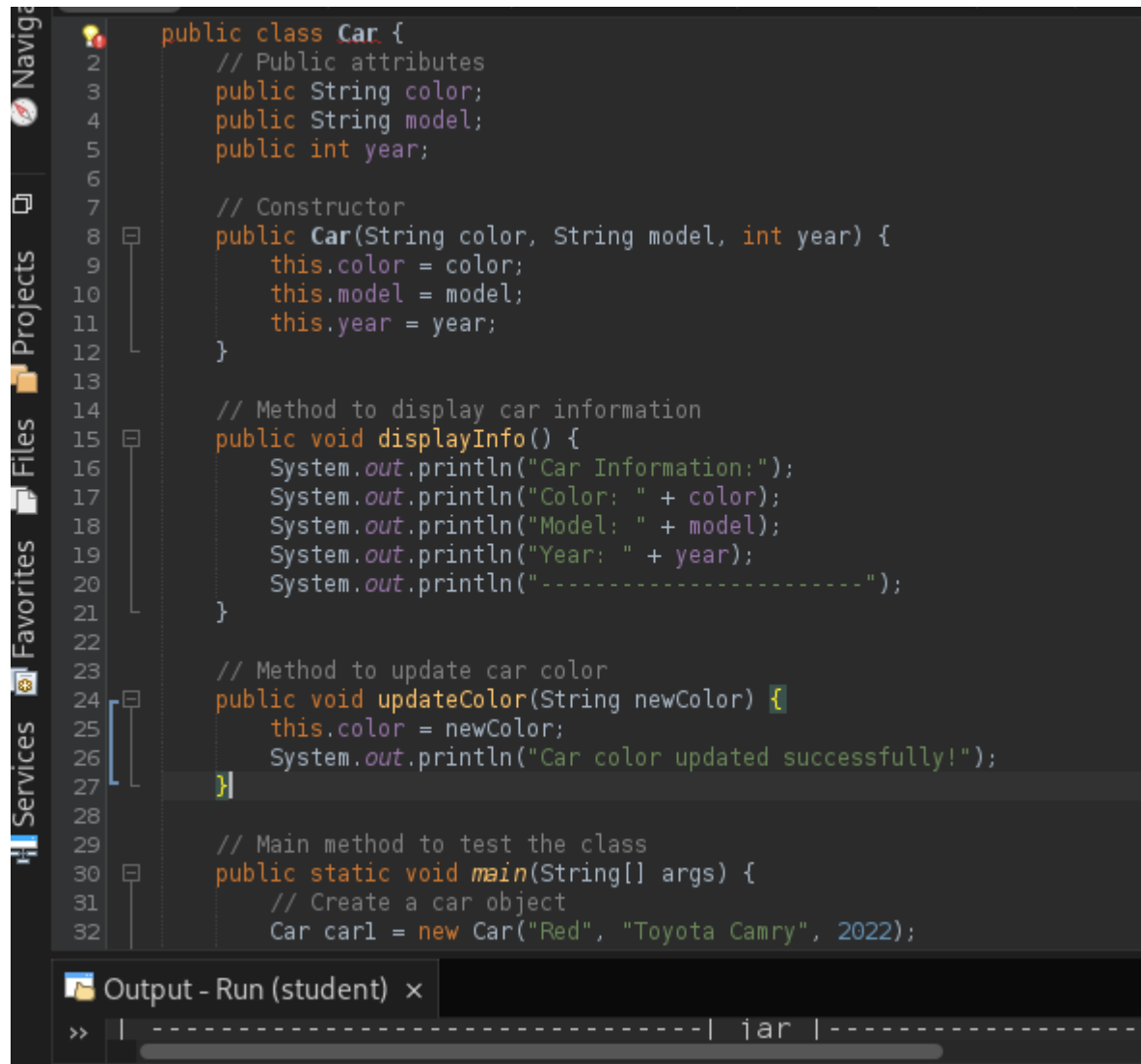
```
11
12
13
14 public class Student {
15     // Public attributes
16     public String name;
17     public int rollNo;
18     public double marks;
19
20     // Constructor
21     public Student(String name, int rollNo, double marks) {
22         this.name = name;
23         this.rollNo = rollNo;
24         this.marks = marks;
25     }
26
27     // Method to display student information
28     public void displayInfo() {
29         System.out.println("Student Information:");
30         System.out.println("Name: " + name);
31         System.out.println("Roll No: " + rollNo);
32         System.out.println("Marks: " + marks);
33         System.out.println("-----");
34     }
35
36     // Method to update marks
37     public void updateMarks(double newMarks) {
38         this.marks = newMarks;
39         System.out.println("Marks updated successfully!");
40     }
41
42     // Main method to test the class
43     public static void main(String[] args) {
44         // Create a student object
45         Student student1 = new Student("Rehan", 102, 92);
46
47         com.mycompany.student.Student >
```

Output:



```
-----[ jar ]-----
--- resources:3.3.1:resources (default-resources) @ student ---
skip non existing resourceDirectory /home/rehan/NetBeansPr
--- compiler:3.13.0:compile (default-compile) @ student ---
Nothing to compile - all classes are up to date.
--- exec:3.5.1:exec (default-cli) @ student ---
Student Information:
Name: Rehan
Roll No: 102
Marks: 92.0
-----
Marks updated successfully!
Student Information:
Name: Rehan
Roll No: 102
Marks: 95.0
-----
BUILD SUCCESS
-----
Total time: 1.318 s
Finished at: 2026-02-10T18:04:33Z
-----
|
```

Task 2:

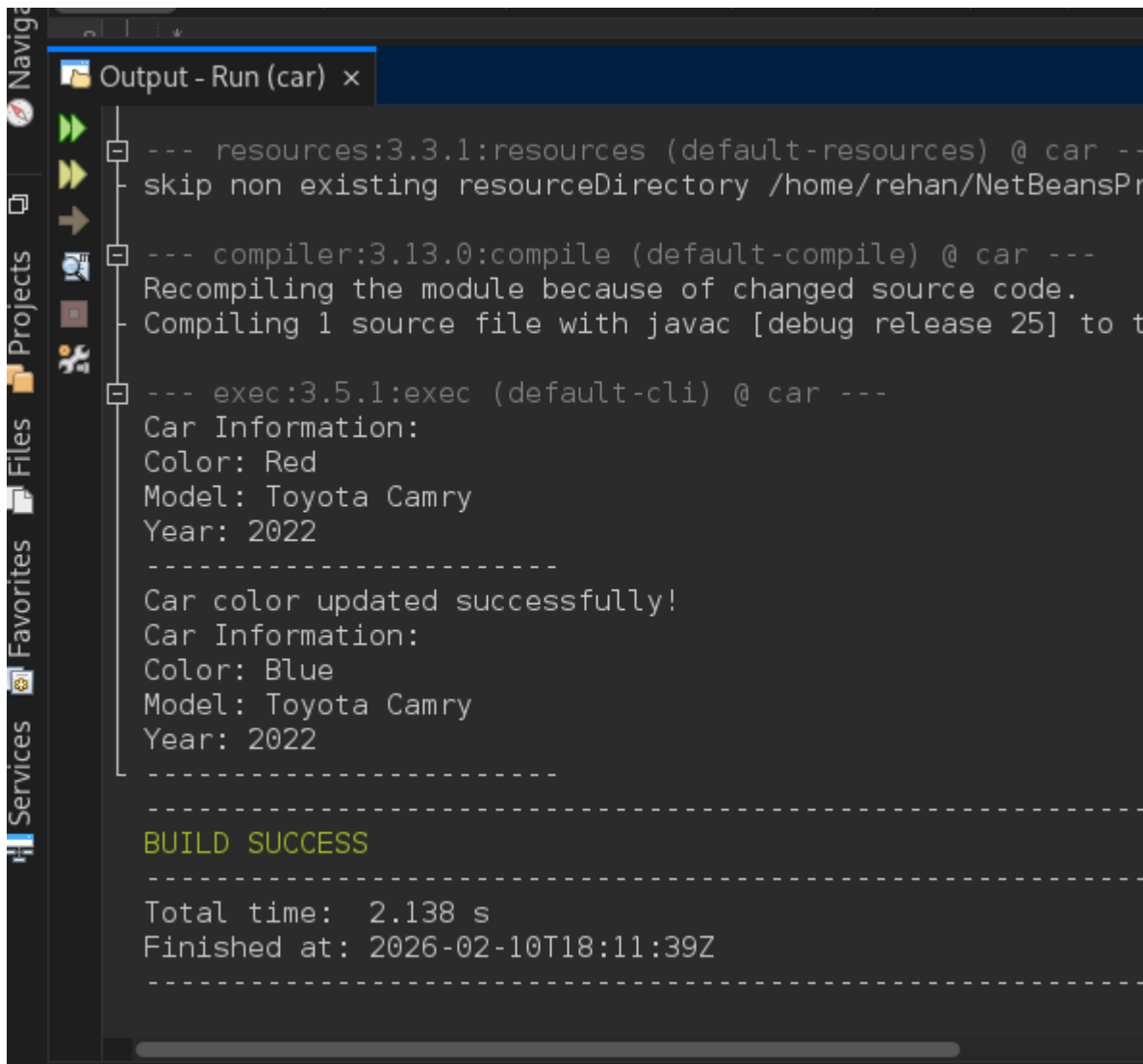


```
1 public class Car {
2     // Public attributes
3     public String color;
4     public String model;
5     public int year;
6
7     // Constructor
8     public Car(String color, String model, int year) {
9         this.color = color;
10        this.model = model;
11        this.year = year;
12    }
13
14    // Method to display car information
15    public void displayInfo() {
16        System.out.println("Car Information:");
17        System.out.println("Color: " + color);
18        System.out.println("Model: " + model);
19        System.out.println("Year: " + year);
20        System.out.println("-----");
21    }
22
23    // Method to update car color
24    public void updateColor(String newColor) {
25        this.color = newColor;
26        System.out.println("Car color updated successfully!");
27    }
28
29    // Main method to test the class
30    public static void main(String[] args) {
31        // Create a car object
32        Car car1 = new Car("Red", "Toyota Camry", 2022);
```

Output - Run (student) x

>> | ----- | | ar | -----

Output:

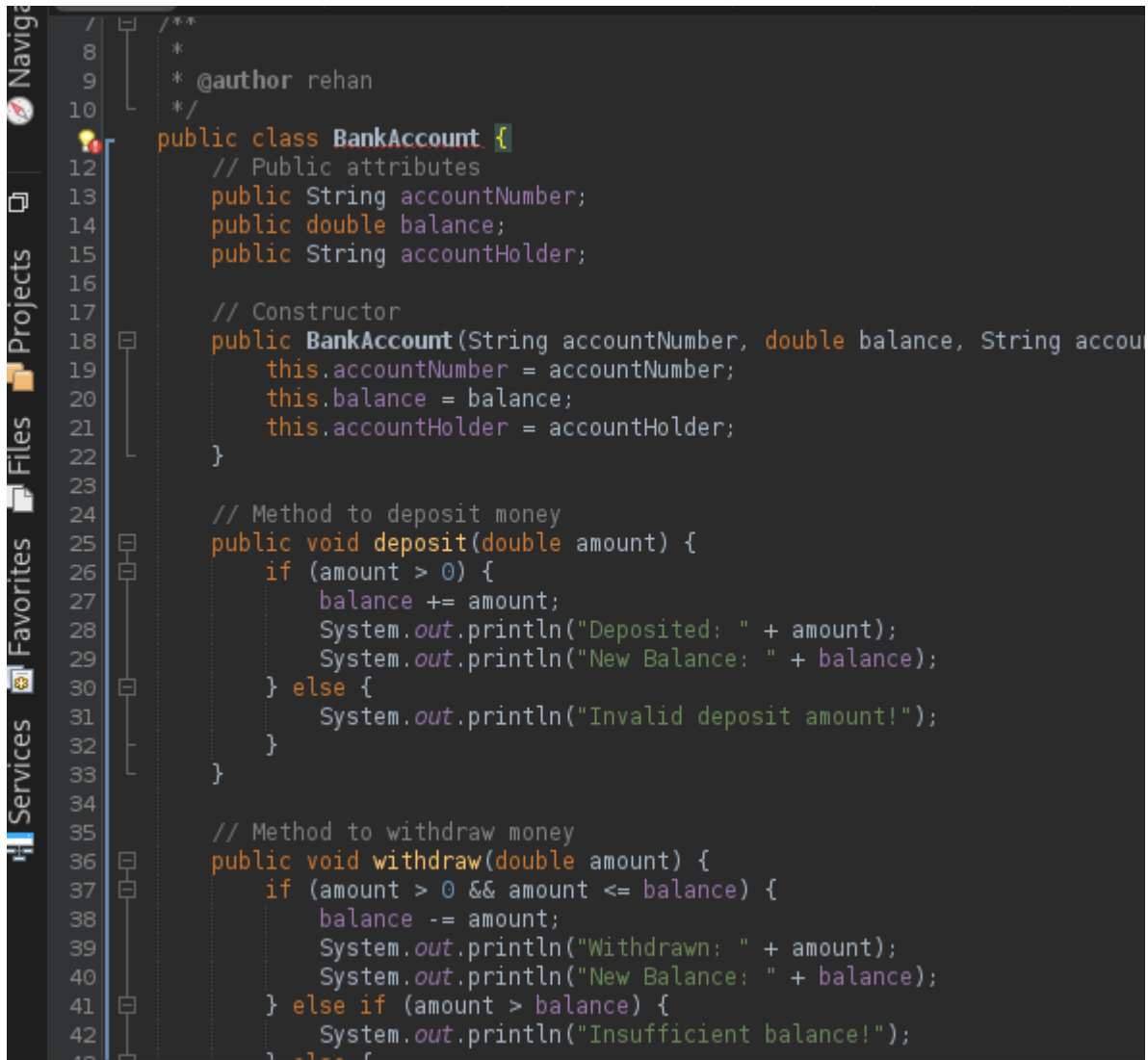


```
Output - Run (car) x
--- resources:3.3.1:resources (default-resources) @ car ---
skip non existing resourceDirectory /home/rehan/NetBeansPr

--- compiler:3.13.0:compile (default-compile) @ car ---
Recompiling the module because of changed source code.
Compiling 1 source file with javac [debug release 25] to t

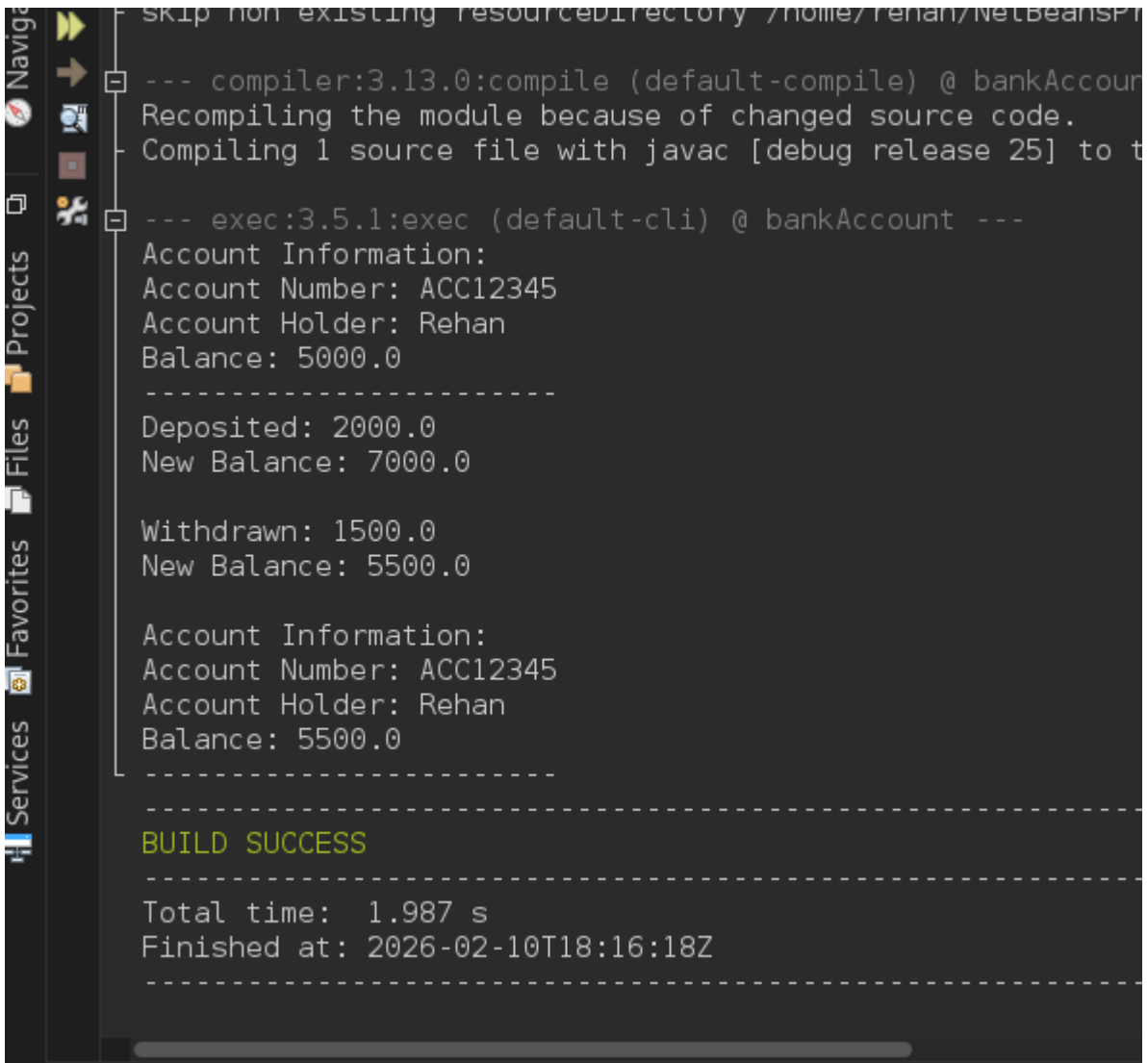
--- exec:3.5.1:exec (default-cli) @ car ---
Car Information:
Color: Red
Model: Toyota Camry
Year: 2022
-----
Car color updated successfully!
Car Information:
Color: Blue
Model: Toyota Camry
Year: 2022
-----
-----
BUILD SUCCESS
-----
Total time: 2.138 s
Finished at: 2026-02-10T18:11:39Z
-----
```

Task 3:

A screenshot of an IDE window showing a Java file named BankAccount.java. The code defines a public class BankAccount with three public attributes: accountNumber (String), balance (double), and accountHolder (String). It includes a constructor that initializes these attributes and two methods: deposit and withdraw. The deposit method checks if the amount is greater than 0 and updates the balance accordingly. The withdraw method checks if the amount is greater than 0 and less than or equal to the current balance, then updates the balance. The IDE interface includes a sidebar with 'Services', 'Favorites', 'Files', 'Projects', and 'Navigator' tabs. The code is syntax-highlighted, and line numbers are visible on the left.

```
8  /**
9  *
10 * @author rehan
11 */
12 public class BankAccount {
13     // Public attributes
14     public String accountNumber;
15     public double balance;
16     public String accountHolder;
17
18     // Constructor
19     public BankAccount(String accountNumber, double balance, String accountHolder) {
20         this.accountNumber = accountNumber;
21         this.balance = balance;
22         this.accountHolder = accountHolder;
23     }
24
25     // Method to deposit money
26     public void deposit(double amount) {
27         if (amount > 0) {
28             balance += amount;
29             System.out.println("Deposited: " + amount);
30             System.out.println("New Balance: " + balance);
31         } else {
32             System.out.println("Invalid deposit amount!");
33         }
34     }
35
36     // Method to withdraw money
37     public void withdraw(double amount) {
38         if (amount > 0 && amount <= balance) {
39             balance -= amount;
40             System.out.println("Withdrawn: " + amount);
41             System.out.println("New Balance: " + balance);
42         } else if (amount > balance) {
43             System.out.println("Insufficient balance!");
44         }
45     }
46 }
```

Output:



```
skip non-existing resourceDirectory /home/rehan/NetBeansPr
--- compiler:3.13.0:compile (default-compile) @ bankAccoun
Recompiling the module because of changed source code.
Compiling 1 source file with javac [debug release 25] to t
--- exec:3.5.1:exec (default-cli) @ bankAccount ---
Account Information:
Account Number: ACC12345
Account Holder: Rehan
Balance: 5000.0
-----
Deposited: 2000.0
New Balance: 7000.0

Withdrawn: 1500.0
New Balance: 5500.0

Account Information:
Account Number: ACC12345
Account Holder: Rehan
Balance: 5500.0
-----
BUILD SUCCESS
-----
Total time: 1.987 s
Finished at: 2026-02-10T18:16:18Z
-----
```

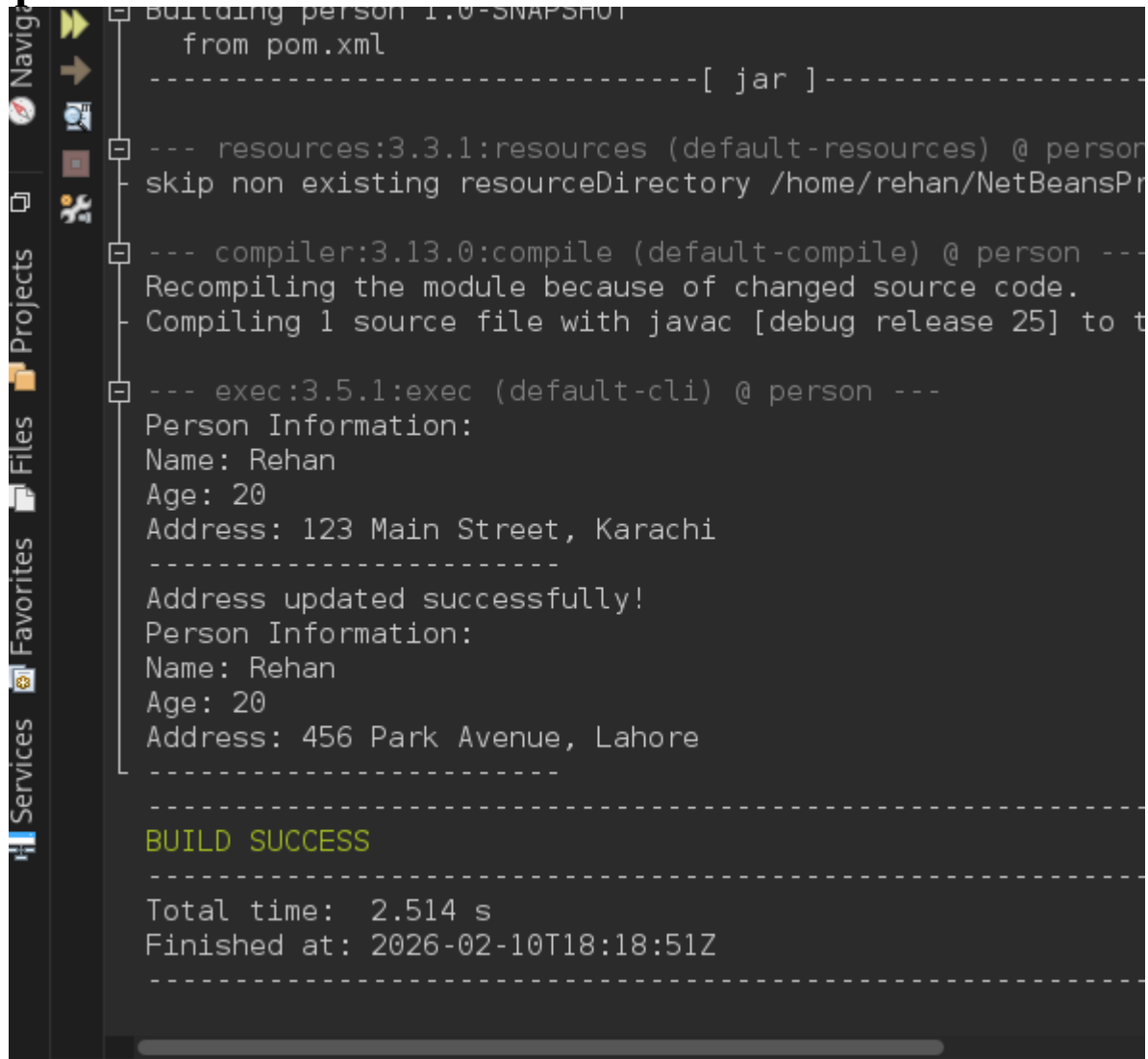
Task 4:

```

8      *
9      * @author rehan
10     */
11     public class Person {
12         // Public attributes
13         public String name;
14         public int age;
15         public String address;
16
17         // Constructor
18         public Person(String name, int age, String address) {
19             this.name = name;
20             this.age = age;
21             this.address = address;
22         }
23
24         // Method to display person information
25         public void displayInfo() {
26             System.out.println("Person Information:");
27             System.out.println("Name: " + name);
28             System.out.println("Age: " + age);
29             System.out.println("Address: " + address);
30             System.out.println("-----");
31         }
32
33         // Method to change address
34         public void changeAddress(String newAddress) {
35             this.address = newAddress;
36             System.out.println("Address updated successfully!");
37         }
38
39         // Main method to test the class
40         public static void main(String[] args) {
41             // Create a person object
42             Person person1 = new Person("Rehan", 20, "123 Main Street, Karachi");
43             // Display person info

```

Output:



```
Building person 1.0-SNAPSHOT
from pom.xml
-----[ jar ]-----
--- resources:3.3.1:resources (default-resources) @ person ---
skip non existing resourceDirectory /home/rehan/NetBeansPr
--- compiler:3.13.0:compile (default-compile) @ person ---
Recompiling the module because of changed source code.
Compiling 1 source file with javac [debug release 25] to t
--- exec:3.5.1:exec (default-cli) @ person ---
Person Information:
Name: Rehan
Age: 20
Address: 123 Main Street, Karachi
-----
Address updated successfully!
Person Information:
Name: Rehan
Age: 20
Address: 456 Park Avenue, Lahore
-----
BUILD SUCCESS
-----
Total time: 2.514 s
Finished at: 2026-02-10T18:51Z
-----
```


Task 5:



```
Source History
9  * @author rehan
10 */
11 public class Rectangle {
12     // Public attributes
13     public double length;
14     public double width;
15
16     // Constructor
17     public Rectangle(double length, double width) {
18         this.length = length;
19         this.width = width;
20     }
21
22     // Method to calculate area
23     public double calculateArea() {
24         return length * width;
25     }
26
27     // Method to display rectangle information
28     public void displayInfo() {
29         System.out.println("Rectangle Information:");
30         System.out.println("Length: " + length);
31         System.out.println("Width: " + width);
32         System.out.println("Area: " + calculateArea());
33         System.out.println("-----");
34     }
35
36     // Main method to test the class
37     public static void main(String[] args) {
38         // Create a rectangle object
39         Rectangle rect1 = new Rectangle(10.0, 5.0);
40
41         // Display rectangle info
42         rect1.displayInfo();
43
44         // Calculate and display area
```

Output

Output:

```
WARNING: sun.misc.Unsafe::staticFieldBase has been c
WARNING: Please consider reporting this to the maint
WARNING: sun.misc.Unsafe::staticFieldBase will be re
Scanning for projects...

-----< com.mycompany:rectangle >---
[+] Building rectangle 1.0-SNAPSHOT
    from pom.xml
-----[ jar ]-----

[+] --- resources:3.3.1:resources (default-resources) @
- skip non existing resourceDirectory /home/rehan/NetB

[+] --- compiler:3.13.0:compile (default-compile) @ rect
- Recompiling the module because of changed source cod
- Compiling 1 source file with javac [debug release 25

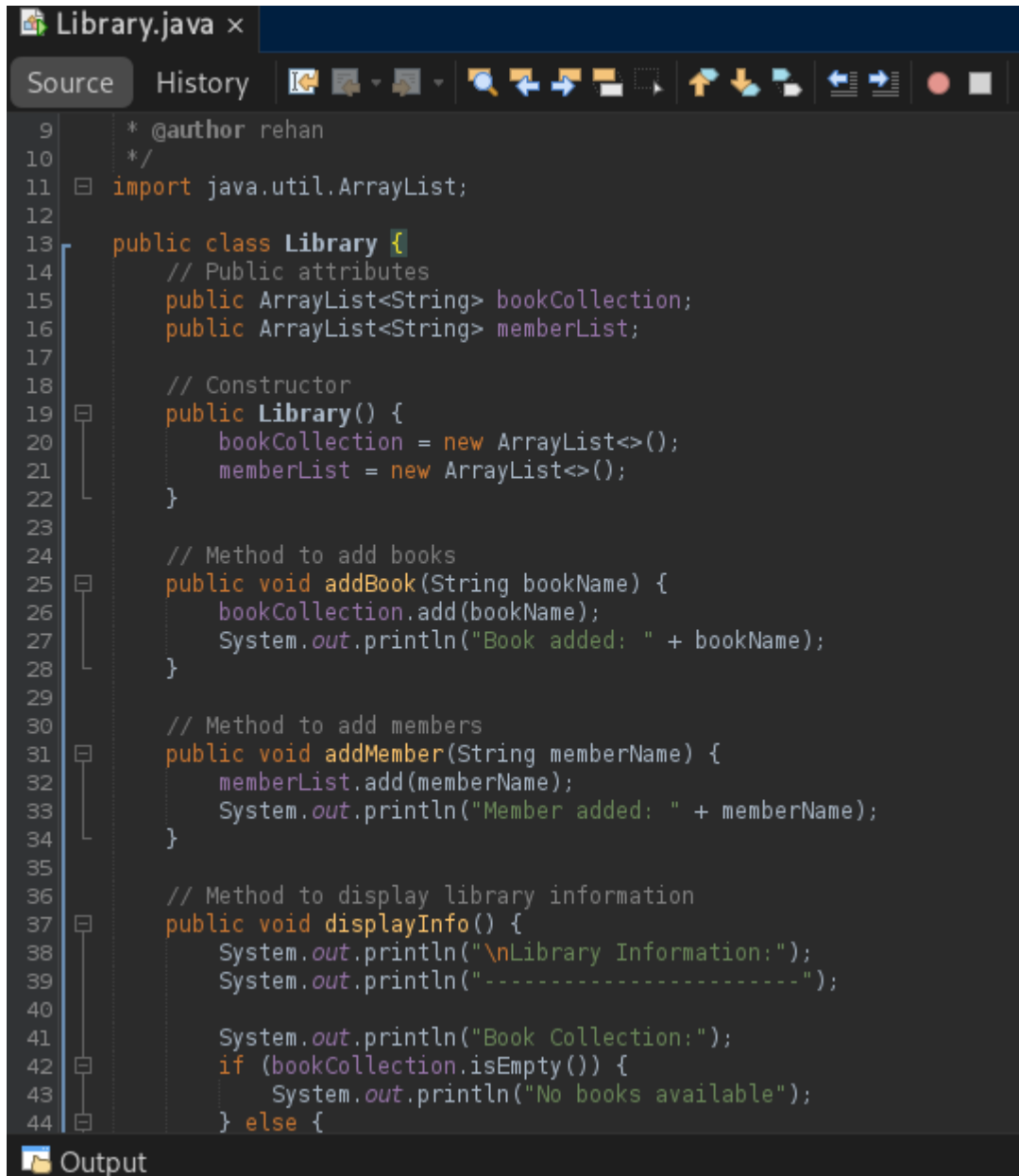
[+] --- exec:3.5.1:exec (default-cli) @ rectangle ---
Rectangle Information:
Length: 10.0
Width: 5.0
Area: 50.0
-----
- The area of the rectangle is: 50.0
-----

BUILD SUCCESS

-----
Total time: 3.708 s
Finished at: 2026-02-11T08:02:50Z
-----
```

Output

Task 6 :



```
9      * @author rehan
10     */
11     import java.util.ArrayList;
12
13     public class Library {
14         // Public attributes
15         public ArrayList<String> bookCollection;
16         public ArrayList<String> memberList;
17
18         // Constructor
19         public Library() {
20             bookCollection = new ArrayList<>();
21             memberList = new ArrayList<>();
22         }
23
24         // Method to add books
25         public void addBook(String bookName) {
26             bookCollection.add(bookName);
27             System.out.println("Book added: " + bookName);
28         }
29
30         // Method to add members
31         public void addMember(String memberName) {
32             memberList.add(memberName);
33             System.out.println("Member added: " + memberName);
34         }
35
36         // Method to display library information
37         public void displayInfo() {
38             System.out.println("\nLibrary Information:");
39             System.out.println("-----");
40
41             System.out.println("Book Collection:");
42             if (bookCollection.isEmpty()) {
43                 System.out.println("No books available");
44             } else {
```

Output

Output:

```
Output - Run (library)
Recompiling the module because of changed source code.
Compiling 1 source file with javac [debug release 25] t
--- exec:3.5.1:exec (default-cli) @ library ---
Book added: Java Programming
Book added: Data Structures
Book added: Database Management

Member added: Rehan
Member added: Ali
Member added: Sara

Library Information:
-----
Book Collection:
1. Java Programming
2. Data Structures
3. Database Management

Member List:
1. Rehan
2. Ali
3. Sara
-----
BUILD SUCCESS
-----
Total time: 4.220 s
Finished at: 2026-02-11T09:03:46Z
-----
|
```

Output